Health and Safety at Work

'Risk Assessment'

Learning aims

You should be able to:

- identify and record hazards in the workplace
- determine the level of risk posed by a hazard using the risk graph
- make recommendations for removal of hazard or reduction of risk

What is risk assessment?

Risk assessment is nothing more than a careful examination of what could cause harm to people, to weigh up whether sufficient precautions have been taken or more should be done to prevent harm.

The purpose is to make sure that no one gets hurt or becomes ill. Accidents and ill-health can ruin individual's lives and affect business too, if output is lost, machinery or equipment is damaged or you have to go to court and you could end up having to pay a fine or go to prison.

Risk assessment of the workplace is a legal requirement.

Starter

- *Who might be affected by accidents at work and in what way???*

There are **five principles of risk assessment** (see *Five Steps to Risk Assessment*, available at www.hse.gov.uk/risk/):

1. Identify hazards
2. Decide who might be harmed, and how
3. Evaluate the risks and decide whether existing precautions are adequate, or more should be done *(use **‘control measures’** to eliminate or prevent ‘hazards’ from becoming accidents’)*
4. Record the findings of the assessment
5. Review the assessment and revise it, if necessary.

Video:



Definitions

- ***Hazard:*** Something that can cause harm, illness or damage to health or property.

- ***Risk:*** The likelihood or chance that harm, illness or damage from a hazard will occur, and the degree of harm (how many people might be affected and how badly).

- ***Control measure:*** Something that reduces or eliminates exposure to a hazard.

Example

- *Table A4.1 shows an extract of a risk assessment for a motor vehicle repair workshop. Note that it explains the standard of safety that is expected to be reached.*

Table A4.1. Five steps to risk assessment

1. Hazard	2. Who might be harmed?	3. Existing controls	4. Standard to be reached	5. Future action(s)
Manual handling In the stores; movement of components.	All employees, particularly those in the stores.	Forklift truck used to move materials into store and take components to workshop. Manual handling still required.	Avoid the risk of injury by removing need for manual handling or provide mechanical aids. If risk cannot be avoided, a more detailed assessment is needed (Manual Handling Operations Regulations)	Supervisor to arrange manual handling training for the staff in the store. More detailed assessment to be carried out

Noise Particularly in body repair work.	All employees, particularly those involved with body work.	Ear defenders must be worn when panel beating. Ear defenders available in vehicle repair for use with certain equipment, for example air saw.	Exposure to noise must be assessed and controlled. Where risk cannot be adequately reduced, hearing protection should be provided.	Supervisors to monitor use of ear defenders
Welding Toxic fumes, sparks, arc-eye.	Employees performing the task. Others nearby.	A range of head and body protection used depending on the type of welding operation. Local exhaust ventilation (LEV) in place.	Exposure to radiation must be prevented. COSHH applies.	Screens to be provided to protect others from radiation. Arrange periodic testing of LEV plant.

… So how do we determine the ***‘level of risk???’*** *…*

Guide lines for determining the level of Risk

Guidelines for determining the level of risk

To decide on what level of risk arises from a hazard, you have to think about:

- the likelihood of an incident and
- the potential effect or severity of an incident.

The **likelihood** can be decided on a scale of 1 to 5.

1. Highly unlikely: not known to occur.
2. Remote possibility: known to occur.
3. Occasional: has happened before.
4. Fairly frequent, some occurrences.
5. Frequent occurrences.

The potential effect or **severity** can also be decided on a scale of 1 to 5.

1. Minor injury.
2. Over three day injury.
3. Temporary incapacity/disease.
4. Permanent incapacity.
5. Fatality.

You use the numbers from these two factors to draw horizontal and vertical lines on the risk graph shown on the next page. You will find the level of risk where the two lines cross.

Action to be taken

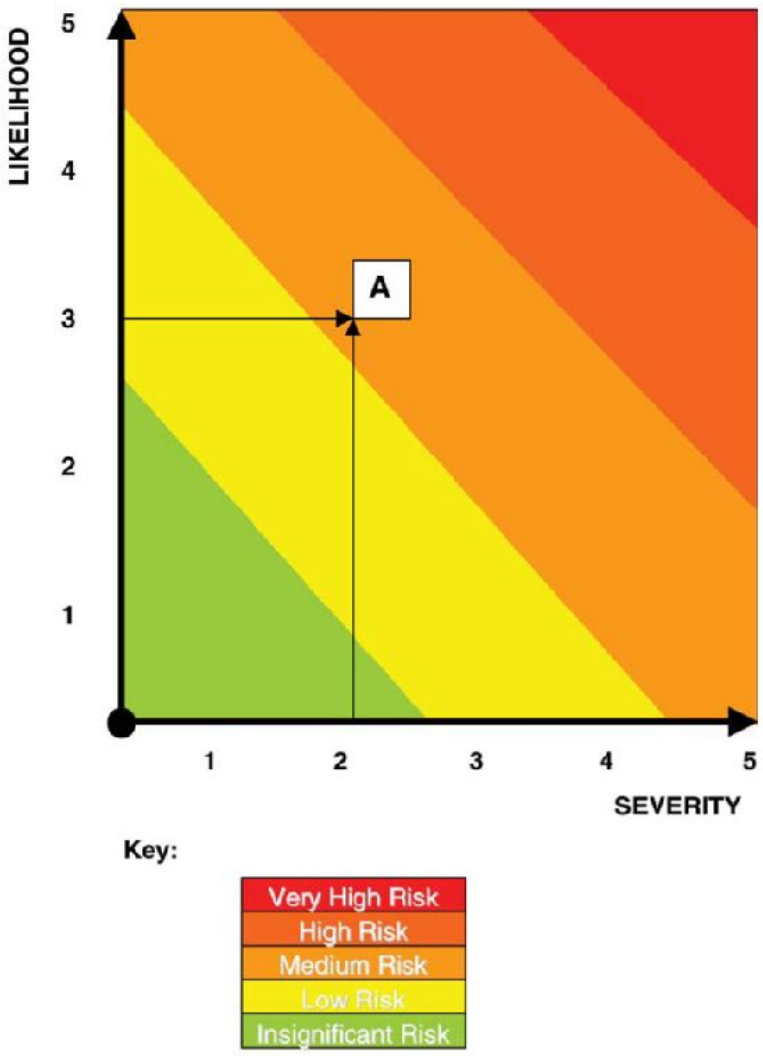
The level of risk that you identify will decide what action should be taken:

Level of risk	Action to be taken
Very high risk	Stop activity. Take immediate action.
High risk	Take action the same day.
Medium risk	Take action within one week.
Low risk	Monitor the situation.
Insignificant risk	No action required.

Example A shows an accident that occasionally happens (**Likelihood number 3**) resulting in an over-three-day injury (**Severity number 2**).

Where the lines cross is identified as a **Medium risk** and action should be taken within one week.

Risk graph



*… Ok, so how do we use the **‘risk graph???’** …*

Example

Using a pillar drill to drill a 10mm diameter hole in a mild steel block 100 x 50 x 25mm

Consider the ***swarf*** – which is the material being removed from the block in order to make the hole

What is the level of risk and severity of an accident occurring during the drilling process involving the swarf?

Ok… *What does this mean???*

Using a pillar drill to drill a 10mm diameter hole in a mild steel block 100 x 50 x 25mm

Consider the ***‘swarf’*** – which is the material being removed from the block in order to make the hole

What is the level of risk and severity of an accident occurring during the drilling process involving the swarf?

First we need to answer the following two questions…

- *What is the ‘hazard’ in this example?*
- *Who might be harmed?*

What is the 'hazard' in this example?

There are a number of possible answers for this question.

*A piece of **'swarf'** could:*

We are going to consider this one…

*A piece of **'swarf'** could be 'ejected' (thrown) from the machine/workpiece – resulting in eye injuries*

Who might be harm?

There are a number of possible answers for this question.

We are going to consider the pillar drill operator

Ok… what about the *level of risk and severity of an accident occurring???*

We now need to answer the following questions…

- *What do we mean by **‘level of risk?’***

  - ***Risk:** The likelihood or chance that harm, illness or damage from a hazard will occur, and the degree of harm (how many people might be affected and how badly).*

This needs our ‘professional’ judgement…
(or use generic risk assessments)

- *This is a ‘remote possibility’ occurrence so we should give a **‘2’***

Ok… what about the *level of risk and severity of an accident occurring???*

We now need to answer the following questions…

- *What do we mean by **‘severity?’***

- ***Severity:*** *The degree of harm that may be caused by the ‘hazard’ becoming an ‘accident’. Ranging from a ‘minor injury’ to a ‘fatality (death)’.*

This needs our ‘professional’ judgement…
(or use generic risk assessments)

- *This could result in ‘temporary incapacity’ so we should give a **‘3’***

We then use the *‘Risk Graph’* in order to determine the:

- *Level of Risk*
- *Action to be taken*

All of this information needs to be recorded on the ‘Risk Assessment’.

Action to be taken

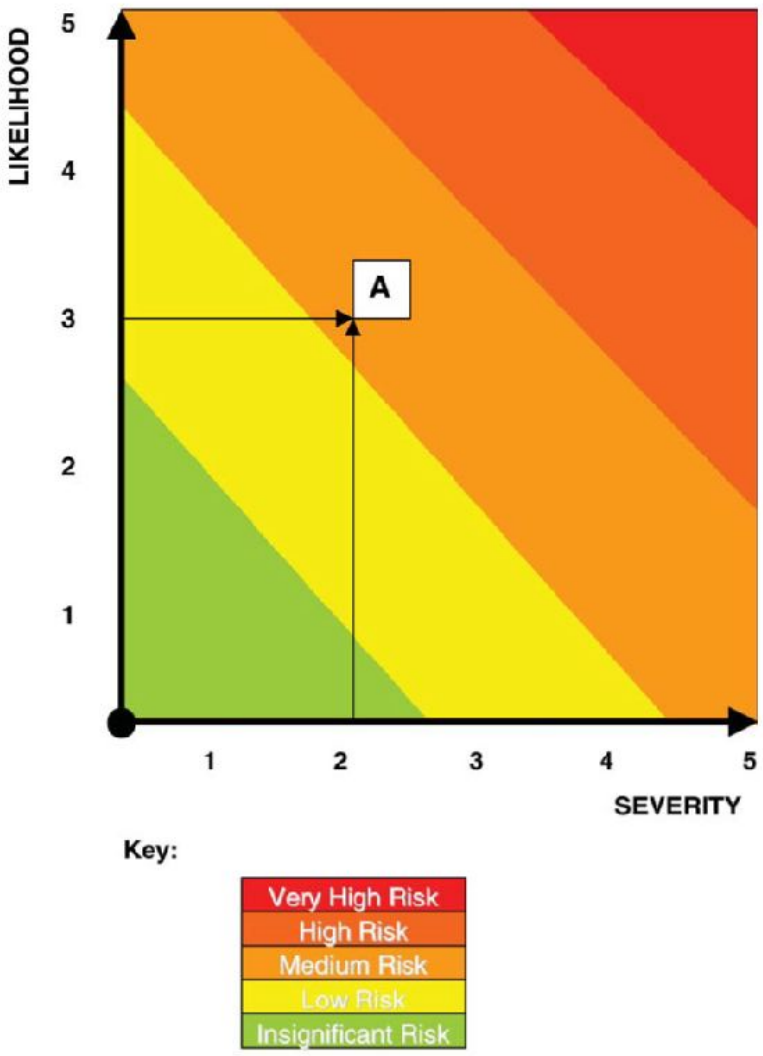
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Example A shows an accident that occasionally happens (**Likelihood number 3**) resulting in an over-three-day injury (**Severity number 2**).

Where the lines cross is identified as a **Medium risk** and action should be taken within one week.

Risk graph



We then use the *‘Risk Graph’* in order to determine the:

- *Level of Risk*

  Medium Risk

- *Action to be taken*

  Take Action within a Week

- *Control Measures to be Implemented*

  - ***All Pillar Drill to be fitted with BS Machine Guards***
  - ***Everyone working in the workshop to wear BS Safety Glasses***

Activity - Carry out a risk assessment Stage 1

Working in small groups, use the ***risk assessment form*** to identify and record:

- any hazards in the AGS workshop (pillar drill)
- who may be exposed to the hazard
- what ***control measures*** are in place – your teacher or trainer can give guidance on how to identify these

Stage 2

- *Assess the **level of risk** of identified hazards by the **Risk Graph**.*

Action to be taken

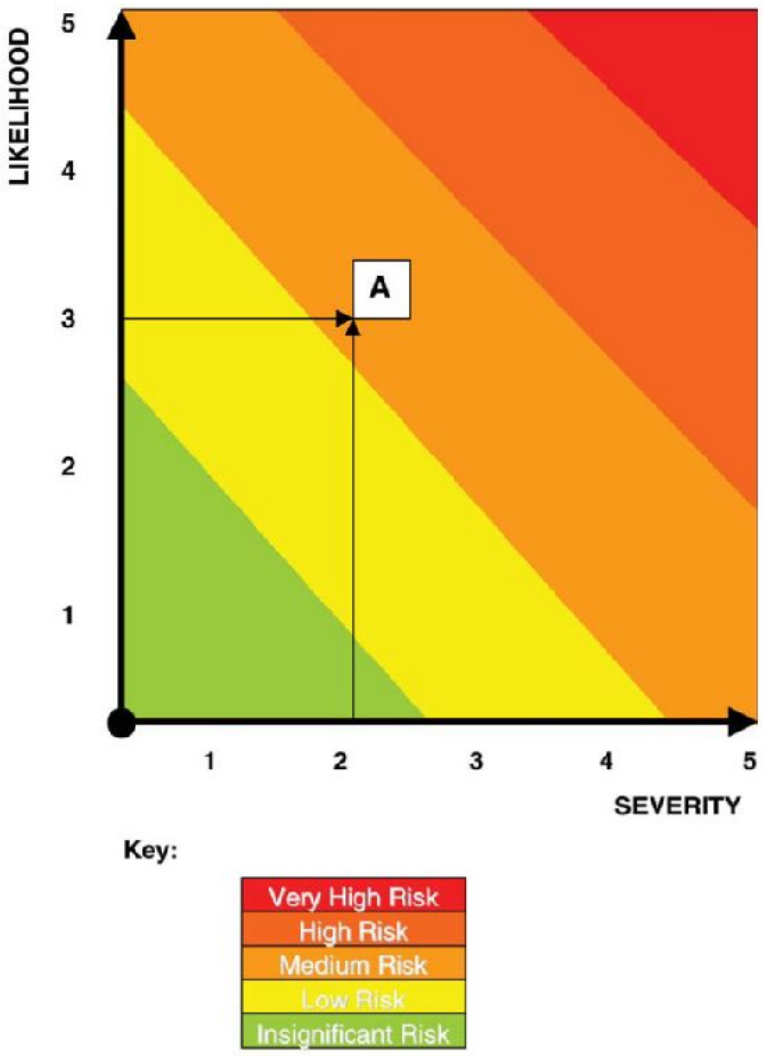
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Example A shows an accident that occasionally happens (**Likelihood number 3**) resulting in an over-three-day injury (**Severity number 2**).

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Risk graph



Stage 3

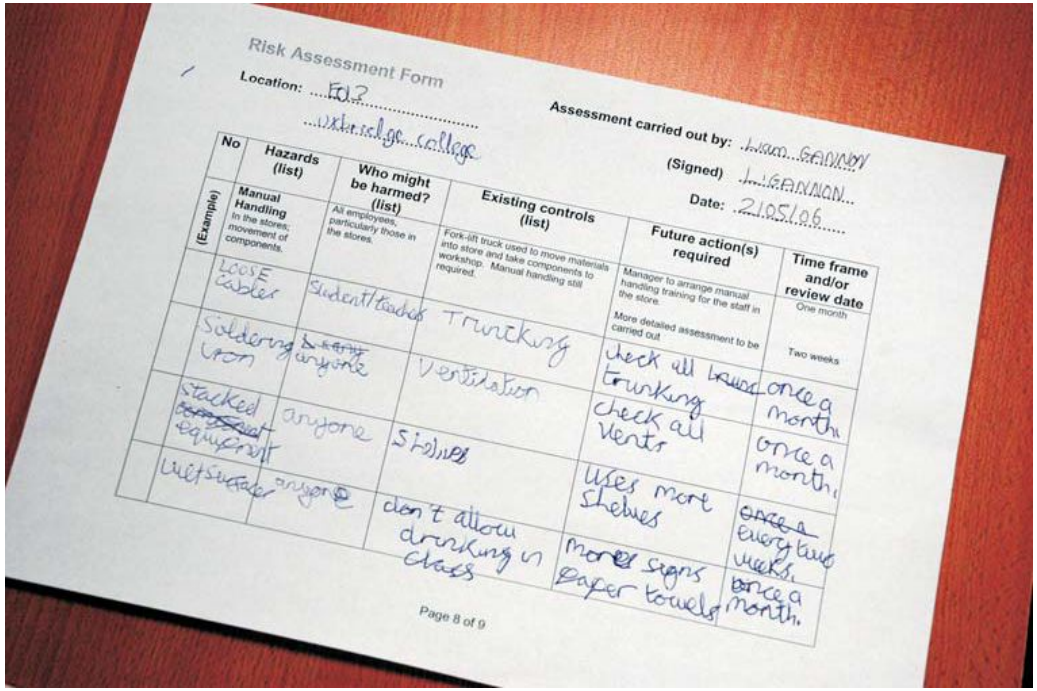
Evaluate the risk. Decide:

- if the existing controls are adequate
- what future actions are recommended
- the time frame to eliminate or minimise the risk.

Record your findings in the ‘future actions’ column of the risk assessment form.

Stage 4

- Provide feedback. Be prepared to discuss your findings and explain your reasoning to the whole class.



Risk Assessment Form

Location: ...ED2... ...Uxbridge college...

Assessment carried out by: L.Gannon

(Signed) L.GANNON

Date: 21/05/06

No	Hazards (list)	Who might be harmed? (list)	Existing controls (list)	Future action(s) required	Time frame and/or review date
(Example)	Manual Handling in the stores: movement of components.	All employees, particularly those in the stores.	Fork-lift truck used to move materials into store and take components to workshop. Manual handling still required.	Manager to arrange manual handling training for the staff in the store.	One month
				More detailed assessment to be carried out	Two weeks
	Loose cables	Student/teacher	Trunking	check all trunking	once a month
	Soldering iron	anyone	Ventilation	check all vents	once a month
	Stacked equipment	anyone	Shelves	Uses more shelves	~~once a~~ every two weeks.
	Wet surface	anyone	don't allow drinking in class	more signs paper towels	~~once a~~ month.

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